

Preface

Who is This Book For?

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Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students ¹. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people ².

Introduction courses to fundamental mathematical topics usually look like this: here're some basic definitions, let's derive some specific results from them - now go apply these in your field.

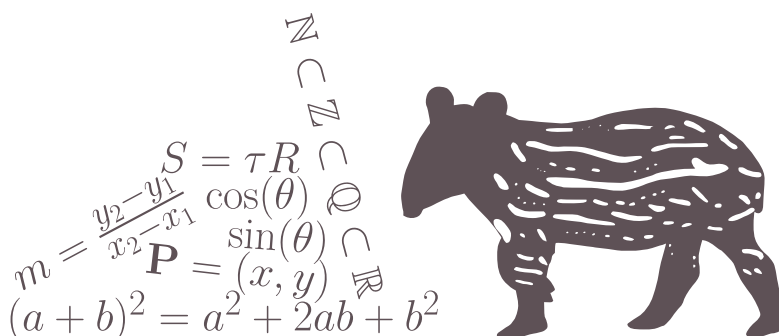
¹and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

²the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

CHAPTER

1

MATHEMATICAL BACKGROUND



$$T(\alpha \vec{x} + \beta \vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} | \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

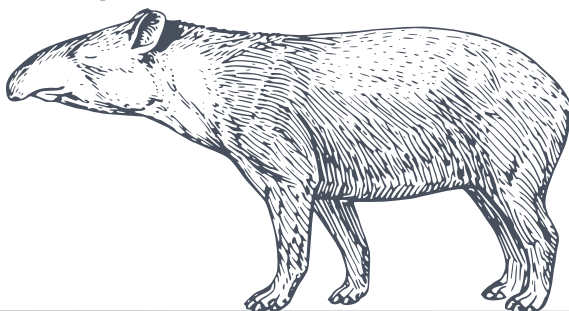
$$(\vec{A} \cdot \vec{B})^T = \vec{B}^T \cdot \vec{A}^T$$

$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A} \vec{x} = \lambda \vec{x}$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_j^i = \langle a^i | b_j \rangle$$



CHAPTER

2

THE BASICS OF LINEAR ALGEBRA

2.1 VECTORS AND VECTOR SPACES

2.1.1 Motivation

2.1.2 Geomteric Properties of Vectors

2.1.3 Geomteric Objects

2.1.4 Vector Spaces

2.1.5 Other Common Coordinate Systems in 2- and 3-Dimensions

2.1.6 Projections and Rejections

2.1.7 Solving the Motivating Problems

2.2 LINEAR TRANSFORMATIONS

2.2.1 Motivation

2.2.2 Linear Transformations in 2-Dimensions

2.2.3 Linear Transformations in 3-Dimensions

2.2.4 The Formal Definition of Linear Transformations

2.2.5 Representing Linear Transformations, Part 1

2.3 MATRICES

2.3.1 Representing Linear Transformations, Part 2

2.3.2 Matrix Representations of 2- and 3-Dimensional Linear Transformations

2.3.3 Types of Matrices

2.3.4 Matrix-Matrix Product

2.3.5 The Determinant

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